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Yeongbin KWON

Education

Columbia University — New York, NY, United States
Master of Science in Civil Engineering and Engineering Mechanics — 2026.08 - 2027.12 (expected)

Seoul National University — Seoul, Republic of Korea
Bachelor of Science in Civil and Environmental Engineering — 2022.03 - 2026.02

- Graduated *summa cum laude* and received the Outstanding Undergraduate Student Commendation from the Korean Society of Coastal Disaster Prevention.
- Bachelor's thesis: *Calibration-Free Stereo Imaging for Coastal Wave Field Reconstruction and Quantification of Wave Breaking*; selected as the Best Bachelor's Thesis of Fall 2025.

Sejong Science High School — Seoul, Republic of Korea
Graduated one year early — 2020.03 - 2022.02

Publications

1. Qin Huang, Moyan Liu, **Yeongbin Kwon**, Sarah Kapnick, and Upmanu Lall. Steering Tropical Cyclones Through Targeted Thermodynamic Perturbations in an AI Weather Model. *JGR: Machine Learning and Computation*, 2026. (*under review*)
2. Qin Huang, Moyan Liu, **Yeongbin Kwon**, and Upmanu Lall. Evaluating the Predictability of Selected Weather Extremes with Aurora, an AI Weather Forecast Model. *Atmosphere*, 17(8):716, 2026. doi:10.3390/atmos17080716

Posters and Presentations

1. **Yeongbin Kwon**, Hyoseob Noh, and Yong Sung Park. Calibration-Free Stereo Imaging for Coastal Wave Field Reconstruction and Quantification of Wave Breaking. *IAHR-APD 2026*, 2026. (poster presentation)
2. Moyan Liu, Qin Huang, Upmanu Lall, **Yeongbin Kwon**, and Hyun-Han Kwon. Weather Jiu-Jitsu: Toward Controlling Atmospheric Rivers with Deep Learning Foundation Models and Adaptive Perturbation. *AGU25 Annual Meeting*, 2025. (poster presentation)
3. Qin Huang, Moyan Liu, Upmanu Lall, **Yeongbin Kwon**, Hyun-Han Kwon, and Yuan Liu. Weather Jiu-Jitsu: Steering Tropical Cyclones with Small Perturbations of the Jet Stream in Experiments with Deep Learning Foundation Models. *AGU25 Annual Meeting*, 2025. (poster presentation)
4. Upmanu Lall, Qin Huang, Moyan Liu, Hyun-Han Kwon, and **Yeongbin Kwon**. Weather Jiu Jitsu: Adaptive Chaos Control with Deep Learning of Atmospheric Dynamics to leverage the power of the atmospheric circulation with nudges to move or defuse storms and avert catastrophic flooding. *AGU25 Annual Meeting*, 2025. (Presented by Upmanu Lall)
5. Upmanu Lall, Moyan Liu, Qin Huang, Hyun-Han Kwon, Adam Nayak, **Yeongbin Kwon**, and Mengjie Zhang. Weather Jiu Jitsu: A 21st century paradigm to leverage nature's power to mitigate some Compound, Concurrent, and Cascading Weather Hazards in a Changing Climate. *AGU25 Annual Meeting*, 2025. (Presented by Upmanu Lall)

Projects

Calibration-Free Stereo Imaging for Coastal Wave Field Reconstruction and Quantification of Wave Breaking
SNU College Undergraduate Independent Research Grant — 2025.05 - 2025.12

- Used stereo imagery to reconstruct coastal wave fields and analyze wave breaking.

Low Impact Development (LID) Strategies for Stormwater Reduction and Reuse on Campus
SNU Integrated Design of Civil Engineering Systems Course — 2025.09-2025.12

- Evaluated low impact development strategies for stormwater reduction and reuse on the SNU campus using SWMM simulations.
- Selected as Outstanding Graduation Project (Top 3).

Internships and Experiences

Researcher | Institute of Engineering Research, SNU 2026.03 - 2026.08

- Conducted post-graduation research at CELAB on identifying rip currents in coastal environments.

Undergraduate Research Intern | Columbia University, New York City 2025.06 - 2025.08

- Used Aurora, a weather foundation model, to study atmospheric dynamics and the effects of targeted perturbations.
- Worked with Prof. Upmanu Lall's group in the Department of Earth and Environmental Engineering; contributed to two AGU25 conference abstracts, a paper published in *Atmosphere*, and a cyclone steering manuscript under review at *JGR: Machine Learning and Computation*.

Community Service

Undergraduate Mentor | Seoul National University Social Responsibility 2025.05 - 2025.12

- Mentored middle school students, focusing on communication and academic support.

Undergraduate Volunteer | Seoul National University Social Responsibility 2025.03 - 2025.06

- Mentored a middle school student from a Bolivian-Korean family for three months, providing academic support.
- Outstanding Volunteer Award recipient (2025.07)

Academic Honors and Awards

Competitive Study Abroad Scholarship (25th Cohort) | Kwanjeong Foundation 2026

Outstanding Undergraduate Student Commendation | Korean Society of Coastal Disaster Prevention 2026.02

Best Bachelor's Thesis Award | SNU 2025.12

Outstanding Graduation Project Award | SNU 2025.12

Merit-Based Tuition Scholarship | SNU Fall 2025; Spring 2024

Outstanding Volunteer Award | Seoul National University Social Responsibility 2025.07

Short-Term Study Abroad Scholarship | SNU Office of International Affairs Spring 2024

Gwanak Scholarship (Merit-Based) | Gwanak Society Fall 2023

Skills

Languages: Korean (native), English (native), Japanese (proficient and fluent).

Programming and Typesetting: LaTeX, `Python`, `MATLAB`, `R`, `C`, `HTML/CSS`, Deep Learning.